

# SHOPPING LIST

Everything to build one, with links. Four hardware items in one Amazon cart, filament from wherever you buy it, and the free arena files. Prices are September 2026 and change; the tax line assumes San Francisco (8.625 %).

**Add all 4 hardware items to Amazon cart**

Opens Amazon with the four items below added. If Amazon shows an error, use the individual links; they're the same products.

## Hardware · one Amazon order

| #                                | ITEM                                                                                                                                                          | WHERE                                                | PRICE   |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|---------|
| 1                                | <b>ESP32 display board, ESP32-2432S028R</b><br>2.8" touch screen "Cheap Yellow Display". Comes with the USB cable and the pigtail leads. Get the single-pack. | <a href="#">Amazon</a><br><a href="#">B0FCXDVBVZ</a> | ~\$14   |
| 2                                | <b>WS2812B LED strip, 5 V, 60 LEDs/m, 1 m</b><br>BTF-LIGHTING, IP30 (no silicone sleeve), black PCB. You use 14 LEDs; the rest is spare.                      | <a href="#">Amazon</a><br><a href="#">B01CDTED80</a> | ~\$9    |
| 3                                | <b>WAGO 221-413 lever nuts, 10-pack</b><br>3-conductor, 24-12 AWG. You use 3. Also at Lowe's / Home Depot in the electrical aisle.                            | <a href="#">Amazon</a><br><a href="#">B06XGYXVXR</a> | ~\$8    |
| 4                                | <b>USB wall charger, 12 W (5 V 2.4 A)</b><br>Amazon Basics one-port USB-A. Any 5 V 2 A+ charger works; the board's cable is USB-A → micro-USB.                | <a href="#">Amazon</a><br><a href="#">B0773J79KC</a> | ~\$7    |
| Sales tax, San Francisco 8.625 % |                                                                                                                                                               |                                                      | ~\$3.30 |

**Hardware total**

**≈ \$41**

Already have a 5 V 2 A phone charger and some lever nuts? Then it's just the board and the strip: about \$25 with tax.

## Filament · Bambu Lab P1S

| ITEM                                                                                                                                         | USED FOR                                                        | WHERE                                              | PRICE   |
|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------|---------|
| <b>Bambu PLA Basic, Silver, 1 kg</b><br>with spool \$18.99 · refill \$15.99<br><a href="#">PRICE READ FROM STORE</a>                         | arena base, lid, plinth ring,<br>top plate, bottom lid (~480 g) | <a href="#">Bambu US store</a>                     | \$18.99 |
| <b>White PLA, any brand, 1 kg</b><br>Bambu PLA Basic White refill \$15.99, or ELEG00 /<br>SUNLU ~\$13, or Bambu Jade White at Best Buy today | beam tube (single-wall vase<br>print), spine, socket (~35 g)    | <a href="#">Amazon</a> · Bambu<br>store · Best Buy | \$13-23 |

\$22.99

### One-spool option

Print everything in white and skip the silver.

all parts

\$13-19

## Free downloads

| ITEM                                                                                                                     | NOTE                                       | WHERE                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Golden 1 Center arena files by Dave Lack</b><br>220 mm base + 220 mm no-hole lid                                      | Free Printables account needed to download | <a href="https://printables.com/model/338758">printables.com/model/338758</a>                       |
| <b>This project's firmware, plinth and beam parts</b><br>OpenSCAD sources, STL exports, PlatformIO firmware, build guide | Open source                                | <a href="https://github.com/zacharygking/light-the-beam">github.com/zacharygking/light-the-beam</a> |

## Optional

| ITEM                                                                          | WHY                                                             | PRICE  |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------|--------|
| <a href="#">USB-C panel-mount jack, 1 ft</a>                                  | Clean USB-C socket on the plinth's back instead of a cable slot | ~\$9   |
| <a href="#">74AHCT125 level shifter</a><br>Adafruit 1787<br><b>PRICE READ</b> | Only if the first LED flickers                                  | \$1.50 |
| <a href="#">Display board 2-pack</a>                                          | A spare board for the next project, ~\$5 more than one          | ~\$20  |

## Tools on hand

- ☐ Scissors or flush cutters
- ☐ Hot glue gun (or super glue)
- ☐ Ruler or calipers
- ☐ Masking tape and a pencil
- ☐ Isopropyl alcohol wipe
- ☐ Phone (WiFi setup) and a computer with USB (flashing, once)

**No soldering iron needed.** The three connections are lever nuts.

## Local pickup near Bayview, San Francisco

| STORE                                                             | GET                                                                | NOTE                                                               |
|-------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Lowe's</b><br>491 Bayshore Blvd · Mon-Sat<br>6a-10p, Sun 7a-8p | WAGO 221-413, USB charger                                          | Ask for the 221-413 (24-12 AWG); the site lists the bigger 221-613 |
| <b>Best Buy</b><br>1717 Harrison St                               | Insignia 12 W USB charger (~\$13), Bambu PLA Basic (\$22.99-24.99) | In-store pickup; White, Gray and Black usually stocked             |

**Home Depot Colma II**  
2 Colma Blvd

WAGO 221-413 10-pack

Backup for Lowe's

The display board isn't stocked in any Bay Area store (Micro Center in Santa Clara carries different WaveShare boards that need other firmware). Order it on Amazon; Prime same-day / next-day covers SF.